

ORBIS AI: AN INTELLIGENT AGENTIC TRAVEL PLANNING AND BOOKING SYSTEM

CS/AI/CY/SE4092 — Final Year Project · FAST NUCES Karachi · Spring 2026

MOTIVATION

- Travel planning today is **deeply fragmented** — users switch between 5+ platforms for flights, hotels, and activities, causing decision fatigue and missed savings.
- Large Language Models (LLMs) have matured enough to **understand complex multi-step intent** and orchestrate downstream tools autonomously.
- No existing consumer product handles **dynamic mid-trip re-planning** — budget changes, flight delays, and availability loss require manual rework.
- RAG-based personalisation enables a system that **learns from each interaction**, improving recommendations progressively over time.

SYSTEM ARCHITECTURE

Architecture Diagram

Figure 1: Orbis AI layered architecture — Next.js 15 frontend communicates with FastAPI backend, which routes through the LangGraph orchestrator to six specialised AI agents backed by Supabase, Redis, and external travel APIs.

MULTI-AGENT ORCHESTRATION

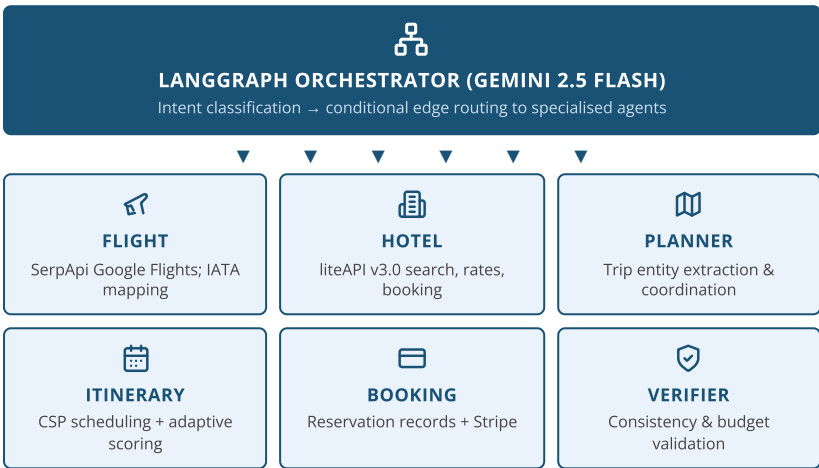


Figure 2: Six ReAct agents share a common AgentState TypedDict across LangGraph edges.

RAG PERSONALISATION PIPELINE

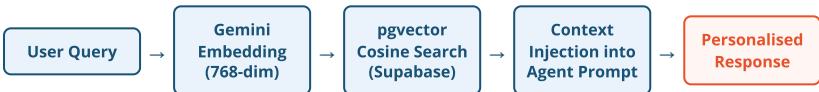
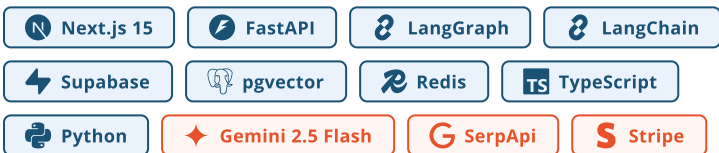


Figure 3: Embeddings cached in Redis (24hr TTL) — reduces Gemini API cost ~60%.

TOOLS & TECHNOLOGIES



CHALLENGES

- Building a reliable **intent routing layer** — ambiguous queries must be classified correctly as the quality of the Orchestrator determines overall usability.
- SSE streaming** is architecturally significant — it constrains how LangGraph emits events and how React state is updated during multi-agent execution.
- External API reliability** — SerpApi and liteAPI can fail; circuit breakers with in-memory fallback must be in place for graceful degradation.
- RAG context quality** — retrieving too much context increases token cost and dilutes relevance; cosine thresholding and caching are essential.
- Constraint satisfaction in real time** — itinerary re-planning must complete in under 3 seconds to maintain the conversational UX.

OUR CONTRIBUTIONS

- Domain-Specialised Multi-Agent Architecture** — six ReAct agents with focused tool suites orchestrated via LangGraph StateGraph, enabling modular and extensible travel AI.
- Live API Integration** — real flight data from SerpApi and hotel availability from liteAPI v3.0; no mocked responses.
- RAG-Based Personalisation** — Gemini-embedded user preferences + travel guides retrieved via pgvector; adaptive scoring weights refined from feedback.
- Dynamic Re-Planning Engine** — constraint propagation handles budget changes, date shifts, and availability loss in real time (<3s).
- Multi-Modal Trip Summariser** — LLM narrative + interactive maps + landmark images + weather overlay generated from structured itinerary JSONB.
- State-of-the-Art Chat Interface** — SSE-streamed responses, structured flight/hotel cards, side-panel artifact display, fully responsive design.

RESULTS

Test Scenario	Result
Flight search & IATA code resolution	✓ Pass
Multi-turn itinerary refinement (history injection)	✓ Pass
Hotel search & pre-booking via liteAPI sandbox	✓ Pass
Dynamic re-planning on budget modification	✓ Pass
Multi-modal summary (narrative + map + weather)	✓ Pass
SSE streaming stability (50-message session)	✓ Pass
Redis circuit breaker + in-memory fallback	✓ Pass
Intent misclassification recovery	~ Partial

PERFORMANCE METRICS

1.5–2.5s First streamed token latency	2–4s Live flight search (SerpApi)
<5ms RAG retrieval (Redis cached)	<3s Dynamic itinerary re-plan



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